
RETINA-SAFE and ECRT: Evidence-Conditioned Hallucination Risk Triage for Medical LLMs

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Abstract

1 Medical large language model (LLM) safety depends on how an answer relates to
2 the available evidence, not on the input scenario alone. A model can safely reject a
3 false premise or defer when evidence is incomplete; the same scenario becomes
4 unsafe when the model asserts a contradicted or unsupported conclusion. We intro-
5 duce RETINA-SAFE, a 12,522-item diabetic retinopathy benchmark that explicitly
6 separates three evidence relations from model-specific response-safety labels. We
7 also introduce ECRT, a two-stage detector that holds the generated response fixed
8 and contrasts the backbone’s logits and hidden states with retinal evidence present
9 (CTX) and removed (NOCTX). Stage 1 detects unsafe responses, and Stage 2 at-
10 tributes contradiction versus evidence-gap risk. Across five patient-disjoint grouped
11 holdouts, ECRT reaches 0.8142 ± 0.0045 balanced accuracy and 0.8845 ± 0.0048
12 AUROC on Llama-3-8B, exceeding FactoScope by 0.0632 balanced accuracy (95%
13 CI: [0.0512, 0.0754]). On Qwen2.5-7B, ECRT reaches 0.8435 ± 0.0041 balanced
14 accuracy and 0.9120 ± 0.0042 AUROC, a 0.0615 gain over FactoScope. At a
15 validation-selected operating point targeting at least 95% unsafe-response recall,
16 the estimated review rate falls from 82.6% with perplexity to 62.8% with ECRT on
17 Llama-3-8B. Text shortcuts, negative controls, counterfactual interventions, and a
18 source-held-out evaluation show that the gain tracks the evidence–response relation
19 within the tested retinal setting.

20 1 Introduction

21 A false premise in a clinical question does not automatically make the model’s response unsafe.
22 A model that rejects the premise acts safely; a model that repeats it as fact does not. The same
23 distinction holds when evidence is missing: deferral is appropriate, unsupported certainty is not. A
24 benchmark that labels the scenario without examining the answer cannot separate these outcomes.

25 Response-level safety changes both the learning target and the review action. Two models can answer
26 the same evidence-conflict or evidence-gap item and receive different labels because their responses
27 use the evidence differently. A useful detector must therefore read the supplied evidence *and* the
28 response that was actually produced. It must also distinguish contradiction from insufficiency: the
29 former calls for correcting a claim, whereas the latter calls for acquiring more evidence.

30 Diabetic retinopathy (DR) grading makes this distinction concrete. DR decisions rely on structured
31 retinal findings, so an answer can fail either by contradicting an observed finding or by claiming more
32 than the findings support. Existing medical hallucination benchmarks primarily evaluate correctness or
33 factuality [Pal et al., 2023, Pandit et al., 2025]. Black-box uncertainty and self-consistency methods
34 score the output without directly testing its dependence on supplied clinical evidence [Manakul
35 et al., 2023, Kadavath et al., 2022]. White-box detectors expose signals in hidden states, logits, or

attention [Azaria and Mitchell, 2023, Chen et al., 2024, He et al., 2023, Chuang et al., 2024], but they do not resolve the response-level distinction between contradiction and insufficient evidence.

We address this gap with RETINA-SAFE and Evidence-Conditioned Risk Triage (ECRT). RETINA-SAFE separates three input relations—E-Align, E-Conflict, and E-Gap—from the safety label of each model response. ECRT follows a simple intervention: keep the response fixed, remove the retinal evidence, and measure what changes inside the backbone. A hierarchy then detects unsafe responses before attributing them to contradiction or evidence insufficiency.

The evaluation follows the same progression. First, ECRT improves Llama-3-8B balanced accuracy over the strongest adapted white-box baseline by 0.0632 (95% CI: [0.0512, 0.0754]). Second, it reduces the estimated review rate by 19.8 percentage points relative to perplexity under the matched validation recall target. Third, its prediction changes in the expected direction on 89.4% of 1,000 counterfactual evidence pairs. Finally, a source excluded from training and threshold selection retains 0.7985 ± 0.0062 balanced accuracy. These results connect the performance gain to evidence-conditioned response triage in structured DR.

Our contributions are:

- **A response-level benchmark.** RETINA-SAFE contains 12,522 DR items with patient, image, source, and construction provenance. Its backbone-specific response labels reach $\kappa = 0.842$ in a 300-item stratified audit.
- **A paired white-box detector.** ECRT contrasts the same response under CTX and NOCTX passes, then combines discrepancy, deviation, and incoherence signals in a safe/unsafe and contradiction/gap hierarchy.
- **A layered evaluation.** Patient-disjoint tests, matched baselines, ablations, negative controls, counterfactuals, five-backbone replication, and source-held-out evaluation examine performance, mechanism, and robustness in sequence.

2 Related Work

Medical hallucination evaluation. Med-HALT and MedHallu evaluate hallucination and reasoning failures in medical LLMs [Pal et al., 2023, Pandit et al., 2025]; ReXTrust studies fine-grained errors in radiology reports [Hardy et al., 2025]. RETINA-SAFE changes the unit of evaluation. It asks whether a generated response is supported, contradicted, or left unresolved by structured retinal evidence, separating the input relation from the safety of the answer.

Hallucination detection. Black-box methods estimate uncertainty, consistency, or reference agreement from outputs [Manakul et al., 2023, Kadavath et al., 2022, Hu et al., 2024, Fadeeva et al., 2023]. White-box approaches use internal states, attention, and related mechanistic signals [Azaria and Mitchell, 2023, Chen et al., 2024, He et al., 2023, Chuang et al., 2024, Sun et al., 2025]. Output uncertainty asks whether a generation is unstable or improbable; reference checking asks whether it agrees with a source; single-pass white-box classification asks whether one internal state predicts factuality. ECRT asks a different question: how does the *same response* change internally when the evidence is removed? Holding decoding constant makes evidence dependence the measured signal and links the detector to contradiction and evidence-gap risk.

3 RETINA-SAFE

3.1 Scenario Relations and Response-Safety Labels

RETINA-SAFE separates the evidence relation in an item from the safety of the response to that item. Each item is $x = (q, o, c, e)$, where q is a clinical question, o contains four options, c is optional context, and e contains image-derived retinal descriptors. The scenario label $s(x)$ is **E-Align** when the evidence supports a unique answer, **E-Conflict** when it rejects a false premise, and **E-Gap** when it is insufficient for a unique answer. These relations follow established DR grading criteria [Wilkinson et al., 2003, Early Treatment Diabetic Retinopathy Study Research Group, 2020, American Diabetes Association Professional Practice Committee, 2024].

Table 1: Response-level labeling policy. The scenario relation constrains the safe action but does not itself determine the response label. “Assert” denotes a definite clinical answer.

Scenario relation	Response behavior	Response-safety label
E-Align	supported gold answer	Safe
E-Align	incorrect or unsupported assertion	Unsafe (Stage 1)
E-Conflict	rejects premise / follows evidence	Safe
E-Conflict	asserts contradicted conclusion	Unsafe-contradiction
E-Gap	abstains or requests evidence	Safe
E-Gap	asserts definite unsupported conclusion	Unsafe-gap

Table 2: RETINA-SAFE scenario relations and response safety for the frozen Llama-3-8B response corpus. Percentages are within scenario. The table shows why scenario relations cannot serve directly as response-safety labels.

Scenario	N	Safe	Unsafe-conflict	Unsafe-gap	Other unsafe
E-Align	1,149	1,082 (94.17%)	24 (2.09%)	18 (1.57%)	25 (2.17%)
E-Conflict	5,107	4,009 (78.50%)	1,098 (21.50%)	0	0
E-Gap	6,266	1,322 (21.10%)	0	4,944 (78.90%)	0
Total	12,522	6,413 (51.21%)	1,122 (8.96%)	4,962 (39.63%)	25 (0.20%)

84 The scenario label describes the input; the response label describes what the model did with it. For
85 each backbone m , we freeze its generated response r_m and parse an action $a_m(x)$: a selected option,
86 an explicit abstention, or an ungradable response. A fixed labeling function compares this action
87 with the gold action, the evidence relation, and the support for any asserted conclusion. The resulting
88 safety label $y_m(x)$ is indexed by the response-producing backbone rather than copied from $s(x)$.

89 Table 1 summarizes the decision logic. Stage 1 assigns every response a safe or unsafe label.
90 Stage 2 assigns evidence-attributable unsafe responses the subtype `unsafe-contradiction` or
91 `unsafe-gap`. Wrong but non-evidence-related and ungradable responses remain unsafe in Stage 1
92 and are excluded from subtype training and scoring; ambiguous responses enter adjudication. Thus,
93 a correct rejection or deferment remains safe even when the input contains conflict or insufficient
94 evidence.

95 3.2 Data Construction, Audit, and Splits

96 RETINA-SAFE v2.0 contains 12,522 items from 4,180 patients, 8,360 eyes, and 10,450 images. The
97 source inventory comprises EyePACS (4,820 records), APTOS 2019 (2,100), Messidor-2 (1,740), and
98 DDR (3,862). Construction first canonicalizes retinal findings into evidence descriptors. Versioned
99 grading rules then inject a rejectable false premise for E-Conflict or remove a descriptor required for
100 a unique answer in E-Gap. The pipeline freezes the gold action before model generation and records
101 the construction rule, source record, evidence template, and question template for every item.

102 The scenario distribution is 1,149 E-Align, 5,107 E-Conflict, and 6,266 E-Gap items. Audit records
103 retain questions, options, evidence descriptors, construction-rule identifiers, and hashed patient/image
104 identifiers; direct identifiers are excluded. Before splitting, we find no exact question–option dupli-
105 cates and no cross-split near duplicates above a MinHash Jaccard threshold of 0.85. The relation
106 label has mutual information 0.0010 with gold-option position and Spearman correlation 0.021 with
107 text length. These checks address two simple leakage routes: answer position and example length.

108 Two DR-domain reviewers independently audited a relation-stratified sample of 300 items and their
109 frozen Llama-3-8B responses while blinded to detector outputs. Table 3 reports five item-level
110 checks. Response-safety labels reached 91.3% raw inter-rater agreement, Cohen’s $\kappa = 0.842$ (95%
111 CI: [0.789, 0.895]), and Gwet’s AC1 of 0.856 (95% CI: [0.805, 0.907]). After adjudication, automatic
112 and human labels agreed on 288/300 responses. The remaining cases follow an adjudication path;
113 Section 5.6 separately analyzes detector failures.

114 The primary protocol uses five 70/15/15 grouped holdouts with seeds 13, 21, 42, 87, and 101. Pa-
115 tients and images never cross train, validation, and test partitions. For seed 13, the split contains

Table 3: Blinded clinical audit of 300 relation-stratified items and frozen Llama-3-8B responses. Counts report agreement with the retinal record or adjudicated reviewer judgment.

Audited component	Agreement	Rate
Evidence descriptor vs. retinal record	295/300	98.3%
Unique gold option	290/300	96.7%
Scenario relation	285/300	95.0%
Expected answer/defer action	283/300	94.3%
Automatic vs. adjudicated response-safety label	288/300	96.0%

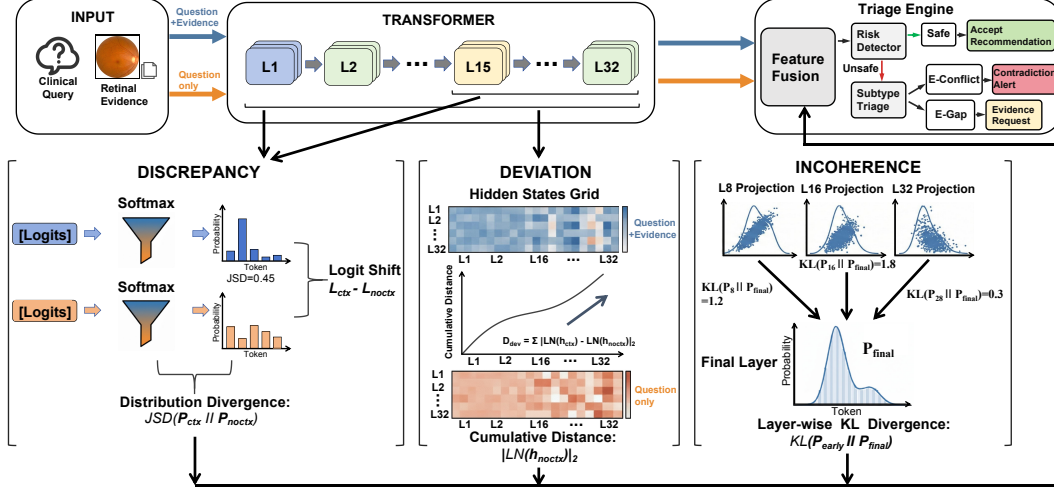


Figure 1: **ECRT overview.** The response generated with retinal evidence is frozen and teacher-forced through the same backbone under CTX and NOCTX conditions. ECRT pools paired logit and hidden-state differences into discrepancy, deviation, and incoherence features. Stage 1 detects unsafe responses; Stage 2 attributes unsafe cases to contradiction or evidence insufficiency.

2,926/627/627 patients and 8,765/1,878/1,879 items. These estimates measure patient-level generalization; prompt-template families may recur across folds. A separate source-held-out experiment excludes every Messidor-2 record from detector training and threshold selection.

4 Evidence-Conditioned Risk Triage

4.1 Paired CTX/NOCTX Probes

ECRT isolates evidence dependence by changing the evidence while holding the response fixed. Let $r = (r_1, \dots, r_T)$ be the response generated from x . The detector teacher-forces r through backbone m twice: CTX uses $(q, o, c, e, r_{<t})$, whereas NOCTX uses $(q, o, c, r_{<t})$. Identical response tokens remove decoding variation from the comparison.

The CTX generation supplies the token sequence for both passes. Explicit masks exclude padding, EOS, and option-control tokens from pooling. The feature extractor never receives the gold answer or response-safety label. Between passes, only e is removed; the question, options, optional context, response prefix, checkpoint, and token positions stay fixed. ECRT therefore measures a paired evidence contrast rather than differences between two independently decoded answers.

At token t and layer ℓ , the passes produce logits $z_t^{\text{ctx}}, z_t^{\text{noctx}}$ and hidden states $h_t^{\text{ctx}, \ell}, h_t^{\text{noctx}, \ell}$. ECRT extracts three signal families:

$$D_t = z_t^{\text{ctx}} - z_t^{\text{noctx}}, \quad V_t^\ell = \left\| \bar{h}_t^{\text{ctx}, \ell} - \bar{h}_t^{\text{noctx}, \ell} \right\|_2, \quad (1)$$

$$I_t = \text{JSD}(\text{softmax}(z_t^{\text{ctx}}) \parallel \text{softmax}(z_t^{\text{noctx}})), \quad (2)$$

where $\bar{h}$ denotes layer-wise normalized states and JSD measures token-distribution incoherence. A fixed masked-mean operator pools non-padding, non-EOS response tokens and selected layers into $\phi(x, r) = [\mathcal{P}(D), \mathcal{P}(V), \mathcal{P}(I)]$.

The three families capture complementary changes. Discrepancy D records how next-token scores move when evidence is removed. Deviation V measures the displacement of normalized hidden representations at selected layers. Incoherence I summarizes the divergence between the predictive distributions. The run manifest fixes pooling, token masks, layer selections, and feature versions within each backbone evaluation. Together, the features measure response dependence on evidence across output and representation spaces; individual dimensions are not assigned clinical meanings.

4.2 Hierarchical Prediction

Stage 1 estimates $p_{\text{unsafe}} = f_1(\phi)$. Stage 2 is trained on subtype-gradable unsafe responses and estimates $p_{\text{gap}|\text{unsafe}} = f_2(\phi)$. The two heads yield the end-to-end probabilities

$$p_{\text{safe}} = 1 - p_{\text{unsafe}}, \quad p_{\text{conflict}} = p_{\text{unsafe}}(1 - p_{\text{gap}|\text{unsafe}}), \quad p_{\text{gap}} = p_{\text{unsafe}}p_{\text{gap}|\text{unsafe}}. \quad (3)$$

Both heads use the same XGBoost recipe. End-to-end three-class Macro-F1 evaluates the full hierarchy on subtype-gradable items. Stage-2 oracle F1 instead bypasses Stage 1 and measures subtype attribution only on gold-unsafe, subtype-gradable responses. All unsafe responses, including other or ambiguous cases, remain in the Stage 1 endpoint.

Training follows the grouped split boundaries. Both heads fit only the training fold, and Stage 1 selects its threshold on validation data to target unsafe-response recall $\geq 95\%$. Stage 2 sees neither safe responses nor unsafe cases without an evidence-attributable subtype. At test time, the Stage 1 gate and Stage 2 conditional probability combine through the equations above. Test labels play no role in fitting or threshold selection.

5 Experiments

5.1 Protocol and Baselines

The experiments build the argument in four steps. **RQ1** tests safe/unsafe discrimination under patient-disjoint evaluation. **RQ2** asks whether the hierarchy improves end-to-end three-class attribution over a flat head. **RQ3** probes whether the paired signal follows evidence instead of text shortcuts or label artifacts. **RQ4** tests whether the result persists across backbones, prompts, and one unseen retinal source.

We evaluate ECRT on backbone-specific response corpora. Llama-3-8B and Qwen2.5-7B form the primary comparison; Llama-2-7B, Llama-2-13B, and Mistral-7B provide additional replications. Within each backbone, every method receives the same frozen responses, response labels, patient splits, and validation-selected operating policy. The baselines span four levels of access. Binary majority and text-only TF-IDF provide lower bounds. Perplexity, mean token entropy, maximum softmax probability, and $P(\text{True})$ [Kadavath et al., 2022] use output uncertainty; SelfCheckGPT [Manakul et al., 2023] adds black-box consistency. Adapted Lookback Lens [Chuang et al., 2024], INSIDE [Chen et al., 2024], and FactoScope [He et al., 2023] provide white-box comparisons. Supervised baselines fit the same training folds and respect the same validation/test boundaries. Flat three-class and signal-family ablations use the ECRT feature space, so they isolate the hierarchy and feature families from the classifier recipe.

Stage-1 balanced accuracy (BA) and AUROC measure discrimination. Specificity at a validation target of unsafe recall $\geq 95\%$ measures review selectivity, and end-to-end three-class Macro-F1 measures full risk attribution. We report mean $\pm$ standard deviation over five grouped holdouts, with the split seed as the replication unit. Method differences use 10,000 paired patient-level bootstrap resamples, keeping responses from each patient together. As an integer check, the seed-13 test set contains 917 unsafe and 962 safe responses; sensitivity 0.9498 and specificity 0.6778 reproduce the reported BA calculation.

Feature extraction uses 16-bit backbones with batch size 1. The XGBoost heads use 160 estimators. Profiling on one NVIDIA RTX 4090 records 70.6 ms total CTX/NOCTX latency per item, 18.4 GB peak memory, and 14.5 GPU-hours for the reported feature-extraction runs.

5.2 Primary Results

Table 4: Primary patient-disjoint results over five grouped holdouts. BA and AUROC are mean $\pm$ std where shown; unadorned entries are five-seed means. Specificity is measured at a validation-selected target of unsafe recall $\geq 95\%$. End-to-end F1 is evaluated on subtype-gradable items and is defined only for methods that output all three response-safety classes. Bold denotes ECRT; paired comparisons are reported in the text.

Backbone	Method	Stage-1 BA $\uparrow$	AUROC $\uparrow$	Specificity $\uparrow$	End-to-end F1 $\uparrow$
Llama-3-8B	Binary majority	0.5000 $\pm$ 0.0000	0.5000 $\pm$ 0.0000	0.0000	–
	Q+E+Response TF-IDF	0.6145 $\pm$ 0.0071	0.6520 $\pm$ 0.0080	0.2790	0.5210
	Perplexity	0.6215 $\pm$ 0.0062	0.6640 $\pm$ 0.0071	0.2930	–
	SelfCheckGPT	0.6580 $\pm$ 0.0072	0.7150 $\pm$ 0.0081	0.3660	–
	FactoScope	0.7510 $\pm$ 0.0051	0.8210 $\pm$ 0.0055	0.5520	–
	Flat three-class	0.7320 $\pm$ 0.0058	0.7950	0.5140	0.7215 $\pm$ 0.0062
	ECRT	0.8142 $\pm$ 0.0045	0.8845 $\pm$ 0.0048	0.6784	0.7834 $\pm$ 0.0049
Qwen2.5-7B	FactoScope	0.7820 $\pm$ 0.0047	0.8530 $\pm$ 0.0050	0.6140	–
	Flat three-class	0.7542	0.8210	0.5580	0.7542
	ECRT	0.8435 $\pm$ 0.0041	0.9120 $\pm$ 0.0042	0.7320	0.8156 $\pm$ 0.0043

ECRT is the strongest Stage 1 detector on both primary backbones. On Llama-3-8B, it improves BA over FactoScope by 0.0632; the paired patient bootstrap gives a 95% CI of [0.0512, 0.0754] and Holm–Bonferroni-adjusted $p < 0.0001$. On Qwen2.5-7B, the gain is 0.0615. Each comparison is made within the corresponding backbone’s frozen-response corpus.

The hierarchy improves the full three-class decision, not just the binary gate. On Llama-3-8B, end-to-end Macro-F1 rises from 0.7215 ± 0.0062 for the flat classifier to 0.7834 ± 0.0049 for ECRT, a gain of 0.0619. With Stage 1 bypassed, the subtype head reaches oracle Macro-F1 of 0.8512 ± 0.0050 on Llama-3-8B and a mean of 0.8765 on Qwen2.5-7B. We report these oracle values separately because they diagnose subtype attribution rather than the end-to-end system.

The gain also changes the review workload. At the validation-selected target of at least 95% unsafe recall, ECRT attains test specificity 0.6784 on Llama-3-8B. With unsafe prevalence 48.8%, it flags approximately 62.8% of items, compared with 82.6% for perplexity at specificity 0.293. Under the matched threshold-selection policy, this difference corresponds to 198 fewer reviews per 1,000 items.

5.3 Do the Gains Depend on the Three Signal Families?

Table 5: Signal-family ablation over five patient-disjoint holdouts. Each entry is Stage-1 BA (mean $\pm$ std). D, V, and I denote discrepancy, deviation, and incoherence.

Features	Llama-3-8B	Qwen2.5-7B
D only	0.7412 $\pm$ 0.0051	0.7685 $\pm$ 0.0048
V only	0.7189 $\pm$ 0.0056	0.7420 $\pm$ 0.0052
I only	0.6945 $\pm$ 0.0062	0.7150 $\pm$ 0.0059
D+V	0.7856 $\pm$ 0.0049	0.8112 $\pm$ 0.0045
D+I	0.7680 $\pm$ 0.0053	0.7945 $\pm$ 0.0049
V+I	0.7521 $\pm$ 0.0057	0.7780 $\pm$ 0.0051
D+V+I	0.8142 $\pm$ 0.0045	0.8435 $\pm$ 0.0041

Every signal family contributes to the final detector. No single family reaches the full representation, and each two-family combination improves on its constituent single-family models. D+V+I performs best on both primary backbones. The next subsection tests the evidence dependence more directly by shuffling and intervening on the evidence itself.

5.4 Shortcuts, Counterfactuals, and Robustness

Surface text alone leaves a substantial gap. TF-IDF over the question, evidence, and response is the strongest text shortcut at 0.6145 ± 0.0071 BA, 0.1997 below ECRT ($p < 0.001$ by paired patient

bootstrap). Question-only, evidence-only, and question-plus-evidence variants reach 0.5318 ± 0.0052 , 0.5482 ± 0.0061 , and 0.5674 ± 0.0058 . Thus, even a text model that observes every surface field does not account for the performance of the paired white-box representation.

The negative controls sharpen this result. CTX-only and NOCTX-only probes underperform the paired representation, so neither pass alone recovers the full signal. Shuffling evidence reduces BA to 0.5210 ± 0.0070 , and permuting labels returns it to 0.5002 ± 0.0035 . The detector therefore depends on the alignment between the evidence and the fixed response, not merely on a static response representation.

Table 6: Evidence and robustness diagnostics. BA results use Llama-3-8B unless a source is named. Counterfactual tests freeze the original response and recompute the target after changing evidence. Each row tests a distinct alternative explanation.

Diagnostic	Quantity	Result	Interpretation within tested scope
CTX-only probe	Stage-1 BA	0.6823 ± 0.0058	Paired contrast adds information
NOCTX-only probe	Stage-1 BA	0.6104 ± 0.0064	Response prior is insufficient
Shuffled evidence	Stage-1 BA	0.5210 ± 0.0070	Correct evidence alignment is required
Permuted labels	Stage-1 BA	0.5002 ± 0.0035	Evaluation returns to chance
Counterfactual pairs ($N = 1,000$)	Prediction-flip accuracy	89.4%	Predictions track changed targets
Consistent $\rightarrow$ conflict	Mean risk-score shift	$+0.482 \pm 0.054$	Risk rises under injected conflict
Five prompt paraphrases	BA standard deviation	0.0068	Limited prompt sensitivity
Option permutations	Absolute BA change	< 0.008	Limited position sensitivity
Messidor-2 held out	BA / AUROC	0.7985/0.8690	Retains signal on one unseen source

Counterfactual pairs provide the strongest relational test. We construct 1,000 pairs that keep the question, options, and original response fixed while changing the evidence and recomputing the response-safety target. ECRT follows the changed target on 89.4% of pairs. When consistent evidence becomes conflicting, the mean risk score rises by 0.482 ± 0.054 across split seeds ($p < 0.001$ by paired patient bootstrap). The detector score therefore tracks controlled changes in the question–evidence–response relation.

5.5 Does the Signal Persist across Backbones and Sources?

The paired signal persists across five backbones. We independently train ECRT with the same feature schema, classifier recipe, and patient-disjoint protocol for each backbone. Every run exceeds the binary majority baseline, and the backbone ordering is consistent across BA, AUROC, and end-to-end Macro-F1 (Table 7). This is an in-backbone replication: each detector trains on response labels produced by the backbone it evaluates.

Table 7: Independent ECRT replication over five patient-disjoint holdouts. Values are mean $\pm$ standard deviation across split seeds. Each row trains a separate detector on responses from the named backbone.

Backbone	Stage-1 BA $\uparrow$	AUROC $\uparrow$	End-to-end F1 $\uparrow$
Llama-2-7B	0.7512 ± 0.0055	0.8120 ± 0.0061	0.7120 ± 0.0058
Llama-2-13B	0.7780 ± 0.0050	0.8412 ± 0.0054	0.7450 ± 0.0052
Mistral-7B	0.7690 ± 0.0052	0.8350 ± 0.0057	0.7380 ± 0.0054
Llama-3-8B	0.8142 ± 0.0045	0.8845 ± 0.0048	0.7834 ± 0.0049
Qwen2.5-7B	0.8435 ± 0.0041	0.9120 ± 0.0042	0.8156 ± 0.0043

We next hold out an entire retinal source. The detector and threshold are fit on EyePACS, APTOS 2019, and DDR, then frozen before evaluation on 1,500 Messidor-2 items that meet pre-specified quality and completeness criteria. ECRT reaches 0.7985 ± 0.0062 BA and 0.8690 ± 0.0068 AUROC. BA decreases by 0.0157 from the primary Llama-3-8B result while remaining well above the binary majority baseline. The result extends the evidence from patient-disjoint splits to one unseen source within structured DR.

5.6 Where Does ECRT Fail?

We review 100 cases sampled evenly from false negatives, false positives, conflict-gap confusions, and low-confidence correct predictions. Two sources dominate this stratified set: missing fine-grained disc or vascular descriptors (32%) and boundary DR grades (28%). Misleading template wording accounts for 20%, tokenizer truncation for 12%, and reviewer-label disagreement at clinical boundaries for 8%. These are proportions of the stratified review set, not population error rates. They point to two practical improvements: preserve descriptors needed for boundary grades, and let the triage policy expose truncation or incomplete evidence instead of forcing a subtype.

6 Discussion

RETINA-SAFE shows why scenario-level labels are too coarse for response safety. In E-Conflict, 78.5% of the frozen Llama-3-8B responses safely reject the false premise or follow the evidence. In E-Gap, 21.1% safely abstain or request more evidence. Conversely, 5.83% of responses to E-Align items are unsafe. Mapping E-Conflict and E-Gap directly to “hallucination” would therefore penalize the behaviors a safety system should preserve, while treating E-Align as automatically safe would miss incorrect or unsupported answers. The response-level formulation changes the prediction target from recognizing a risky prompt to judging how a particular answer uses the evidence.

The diagnostic pattern also explains why the paired intervention helps. A CTX-only probe reaches 0.6823 ± 0.0058 BA and a NOCTX-only probe reaches 0.6104 ± 0.0064 , whereas their paired contrast reaches 0.8142 ± 0.0045 . Shuffling the evidence collapses BA to 0.5210 ± 0.0070 , and controlled counterfactual changes produce the expected prediction flip in 89.4% of pairs. The useful signal is therefore not simply a strong response representation or an evidence representation in isolation. It lies in how the fixed response changes across the two evidence conditions.

The hierarchy turns that relational signal into a review policy. Stage 1 answers whether a response needs review; Stage 2 asks why, separating a claim that conflicts with observed findings from one that outruns incomplete evidence. This decomposition improves end-to-end Macro-F1 by 0.0619 over a flat three-class head. At the matched validation recall target, it also corresponds to 198 fewer flagged items per 1,000 than perplexity. The error review suggests a natural extension: route missing descriptors and truncated contexts to an explicit evidence-incomplete state before attempting subtype attribution. In this form, ECRT is best understood as a triage layer that connects detection to the next review action.

7 Limitations and Ethical Considerations

The present evidence covers response triage from structured, image-derived retinal descriptors. It does not evaluate end-to-end fundus-image diagnosis. The benchmark focuses on DR, the primary split isolates patients but not prompt-template families, and the source-held-out test covers one retinal source. Cross-disease, cross-institution, multilingual, and prospective performance remain open questions. The controlled text interventions establish detector sensitivity to the evidence-response relation; they do not establish causal clinical reasoning by the backbone.

ECRT requires access to logits and hidden states and adds a second teacher-forced pass, limiting its use with closed models. Automatic response labels achieve high agreement in the 300-item audit, while ambiguous and boundary cases still require expert adjudication.

ECRT is a research detector, not a clinical device or an authorization mechanism for treatment. The construction pipeline removes direct identifiers and retains only hashed patient/image identifiers in audit files. Clinical deployment would require prospective validation, demographic and device-stratified fairness analysis, institution-specific governance, and monitoring for shifts in prevalence and model behavior.

8 Conclusion

Medical LLM safety is a relation between an answer and the evidence available for that answer. RETINA-SAFE makes this relation explicit by separating input scenarios from response-level safety

labels. ECRT measures it by comparing the same response with retinal evidence present and removed, then uses the paired signal to detect and attribute unsafe answers. Across five patient-disjoint holdouts, the hierarchy improves balanced accuracy and end-to-end three-class Macro-F1 over matched baselines. Text controls, counterfactual interventions, five-backbone replication, and an unseen retinal source all reinforce the same result: evidence dependence is a measurable signal for response triage. This turns contradiction and unsupported certainty into a concrete safety target for medical LLMs, ready for extension to direct multimodal and prospective clinical evaluation.

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A Additional Data and Label Details

The primary seed-13 split contains 2,926/627/627 patients and 8,765/1,878/1,879 items in train/validation/test. Patient and image overlap is zero. The response-label schema records the backbone, selected option, abstention status, answer correctness, evidence support, unsafe binary label, unsafe subtype, annotators, and guideline version. Template-family identifiers are retained in the split manifest, but the reported primary protocol does not isolate them; its estimates should therefore not be read as unseen-template performance.

B Full Shortcut Controls

Table 8: Llama-3-8B Stage-1 shortcut controls on the patient-disjoint protocol. Values are mean $\pm$ std over five holdouts.

ID	Input and method	BA
S0	Binary majority (always unsafe)	0.5000 $\pm$ 0.0000
S1	Length/position/punctuation + logistic regression	0.5124 $\pm$ 0.0041
S2	Question-only TF-IDF + XGBoost	0.5318 $\pm$ 0.0052
S3	Evidence-only TF-IDF + XGBoost	0.5482 $\pm$ 0.0061
S4	Options-only TF-IDF + XGBoost	0.5189 $\pm$ 0.0038
S5	Question+evidence TF-IDF + XGBoost	0.5674 $\pm$ 0.0058
S7	Response-only TF-IDF + logistic regression	0.5812 $\pm$ 0.0064
S8	Question+evidence+response TF-IDF + XGBoost	0.6145 $\pm$ 0.0071

S6 is reserved for a non-deployable template diagnostic and is excluded from the detector comparison.

358 C Complete Baseline Matrix

Table 9: Complete Stage-1 baseline comparison under the same frozen-response, patient-disjoint protocol. Values are mean $\pm$ standard deviation over five grouped holdouts.

Category	Method	Llama-3-8B		Qwen2.5-7B	
		BA $\uparrow$	AUROC $\uparrow$	BA $\uparrow$	AUROC $\uparrow$
Lower bound	Binary majority	0.5000 $\pm$ 0.0000	0.5000 $\pm$ 0.0000	0.5000 $\pm$ 0.0000	0.5000 $\pm$ 0.0000
Text	Q+E+Response TF-IDF	0.6145 $\pm$ 0.0071	0.6520 $\pm$ 0.0080	0.6320 $\pm$ 0.0065	0.6712 $\pm$ 0.0074
Uncertainty	Perplexity	0.6215 $\pm$ 0.0062	0.6640 $\pm$ 0.0071	0.6480 $\pm$ 0.0055	0.6890 $\pm$ 0.0062
Uncertainty	Mean token entropy	0.6190 $\pm$ 0.0065	0.6615 $\pm$ 0.0075	0.6450 $\pm$ 0.0058	0.6850 $\pm$ 0.0065
Uncertainty	Maximum softmax probability	0.6180 $\pm$ 0.0059	0.6590 $\pm$ 0.0068	0.6412 $\pm$ 0.0058	0.6810 $\pm$ 0.0061
Self-report	P(True)	0.6350 $\pm$ 0.0068	0.6820 $\pm$ 0.0074	0.6625 $\pm$ 0.0061	0.7120 $\pm$ 0.0069
Black-box	SelfCheckGPT ($N = 5$)	0.6580 $\pm$ 0.0072	0.7150 $\pm$ 0.0081	0.6890 $\pm$ 0.0065	0.7480 $\pm$ 0.0072
White-box	Lookback Lens	0.7290 $\pm$ 0.0058	0.7920 $\pm$ 0.0064	0.7580 $\pm$ 0.0052	0.8240 $\pm$ 0.0058
White-box	INSIDE (adapted)	0.7425 $\pm$ 0.0054	0.8120 $\pm$ 0.0060	0.7710 $\pm$ 0.0049	0.8410 $\pm$ 0.0053
White-box	FactoScope	0.7510 $\pm$ 0.0051	0.8210 $\pm$ 0.0055	0.7820 $\pm$ 0.0047	0.8530 $\pm$ 0.0050
Ours	ECRT	0.8142 $\pm$ 0.0045	0.8845 $\pm$ 0.0048	0.8435 $\pm$ 0.0041	0.9120 $\pm$ 0.0042

359 D Evidence Perturbation Detail

Table 10: Llama-3-8B evidence perturbation analysis with the original response frozen. The response-safety target is recomputed after each intervention. BA measures classification under the changed target and need not vary monotonically; the mean risk score measures the detector response to evidence degradation or conflict.

Level	Evidence intervention	Stage-1 BA	Mean risk score
0	Full evidence	0.8142	0.214
1	Remove one non-critical descriptor	0.7915	0.245
2	Remove a grade-defining descriptor	0.7320	0.582
3	Inject explicit conflicting evidence	0.8410	0.895
4	Remove all retinal evidence	0.5110	0.942

360 E Reproducibility Summary

361 The reproducibility protocol fixes split identifiers for seeds 13, 21, 42, 87, and 101. It records
362 model and prompt configurations, response hashes, feature-extractor versions, per-item predictions,
363 bootstrap scripts, and a run manifest linking each table entry to its configuration and commit. CTX
364 and NOCTX teacher-force the same CTX-generated response tokens, use explicit token masks, and
365 apply layer-wise ℓ_2 normalization before masked mean pooling.